

Critical Path Method (CPM) Analysis Report

Project: Vendora Marketplace Development

Project Summary:

Total Duration: 19 days

Total Tasks: 37

Critical Path:

1.1.1 -> 1.2.1 -> 1.2.3 -> 1.3.1 -> 1.3.3 -> 2.1.1 -> 2.1.3 -> 2.2.1 -> 2.2.4 -> 2.3.1 -> 2.3.3 -> 2.3.4 -> 2.4.1 -> 2.4.2 -> 3.3.1 -> 5.1.1 -> 5.1.2 -> 5.1.3 -> 5.3.2 -> 6.3.1

Notes on Methodology:

1. Forward Pass: Calculated Early Start (ES) and Early Finish (EF).
2. Backward Pass: Calculated Late Start (LS) and Late Finish (LF).
3. Slack Calculation: $Slack = LS - ES$. If $Slack = 0$, the task is on the Critical Path.
4. Diagram Format: Each node contains [ES|Dur|EF], [ID & Name], and [LS|Slack|LF].

Task Analysis Table

Task ID	Task Name	Duration	ES	EF	LS	LF	Slack	Critical
1.1.1	Secure signatures	1	0	1	0	1	0	True
1.1.2	Stakeholder strateg	1	1	2	18	19	17	False
1.2.1	Initialize GitHub	1	1	2	1	2	0	True
1.2.2	Setup ClickUp	1	1	2	18	19	17	False
1.2.3	Configure dev env	1	2	3	2	3	0	True
1.3.1	Finalize architecture	1	3	4	3	4	0	True
1.3.2	DB schema	1	4	5	18	19	14	False
1.3.3	API structure	1	4	5	4	5	0	True
2.1.1	Login	1	5	6	5	6	0	True
2.1.2	Logout	1	6	7	18	19	12	False
2.1.3	Role auth	1	6	7	6	7	0	True
2.2.1	CRUD listings	1	7	8	7	8	0	True
2.2.2	Metadata fields	1	8	9	16	17	8	False
2.2.3	Stock status	1	8	9	18	19	10	False
2.2.4	Image upload	1	8	9	8	9	0	True
2.3.1	Product catalog	1	9	10	9	10	0	True
2.3.2	Pagination	1	10	11	18	19	8	False
2.3.3	Product page	1	10	11	10	11	0	True
2.3.4	Shopping cart	1	11	12	11	12	0	True
2.4.1	Checkout	1	12	13	12	13	0	True
2.4.2	Order storage	1	13	14	13	14	0	True
2.4.3	Success screen	1	13	14	18	19	5	False
3.1.1	AI API integration	1	9	10	17	18	8	False
3.1.2	Manual override	1	10	11	18	19	8	False
3.1.3	Loading indicator	1	10	11	18	19	8	False
3.2.1	Recommendation eng	1	10	11	18	19	8	False
3.3.1	Payment gateway	1	14	15	14	15	0	True
4.1.1	Dashboard UI	1	8	9	12	13	4	False
4.1.2	Order tracking	1	9	10	13	14	4	False
4.1.3	Status update	1	10	11	14	15	4	False
5.1.1	End-to-end testing	1	15	16	15	16	0	True
5.1.2	Fix API limits	1	16	17	16	17	0	True
5.1.3	AI stability	1	16	17	16	17	0	True
5.2.1	Storage audit	1	16	17	18	19	2	False
5.2.2	Timeout optimization	1	16	17	18	19	2	False
5.3.2	Final validation	1	17	18	17	18	0	True
6.3.1	Deploy to Vercel	1	18	19	18	19	0	True

CPM Network Diagram (AON)

